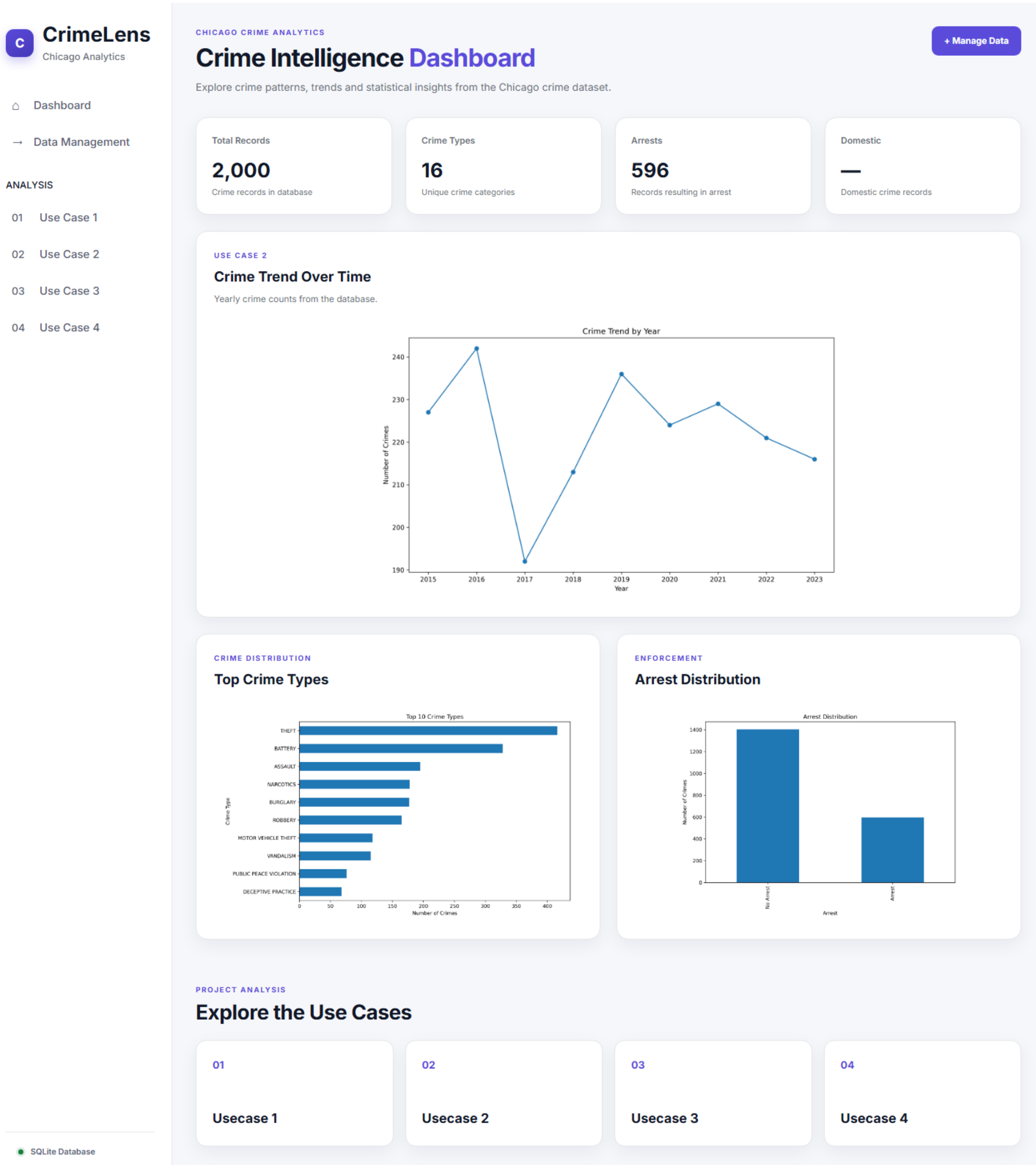
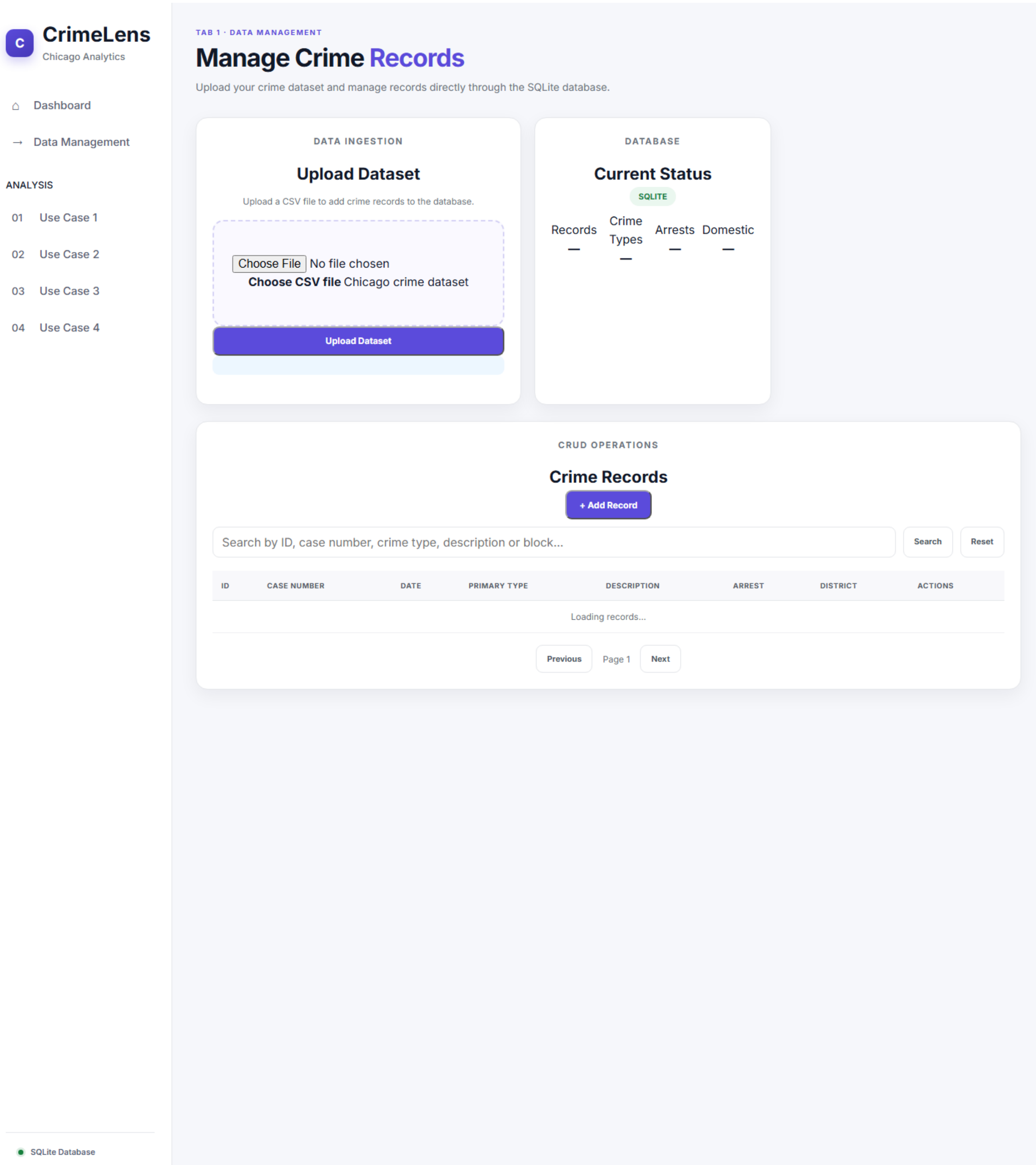
Chicago Crime Analytics — Frontend Screenshots

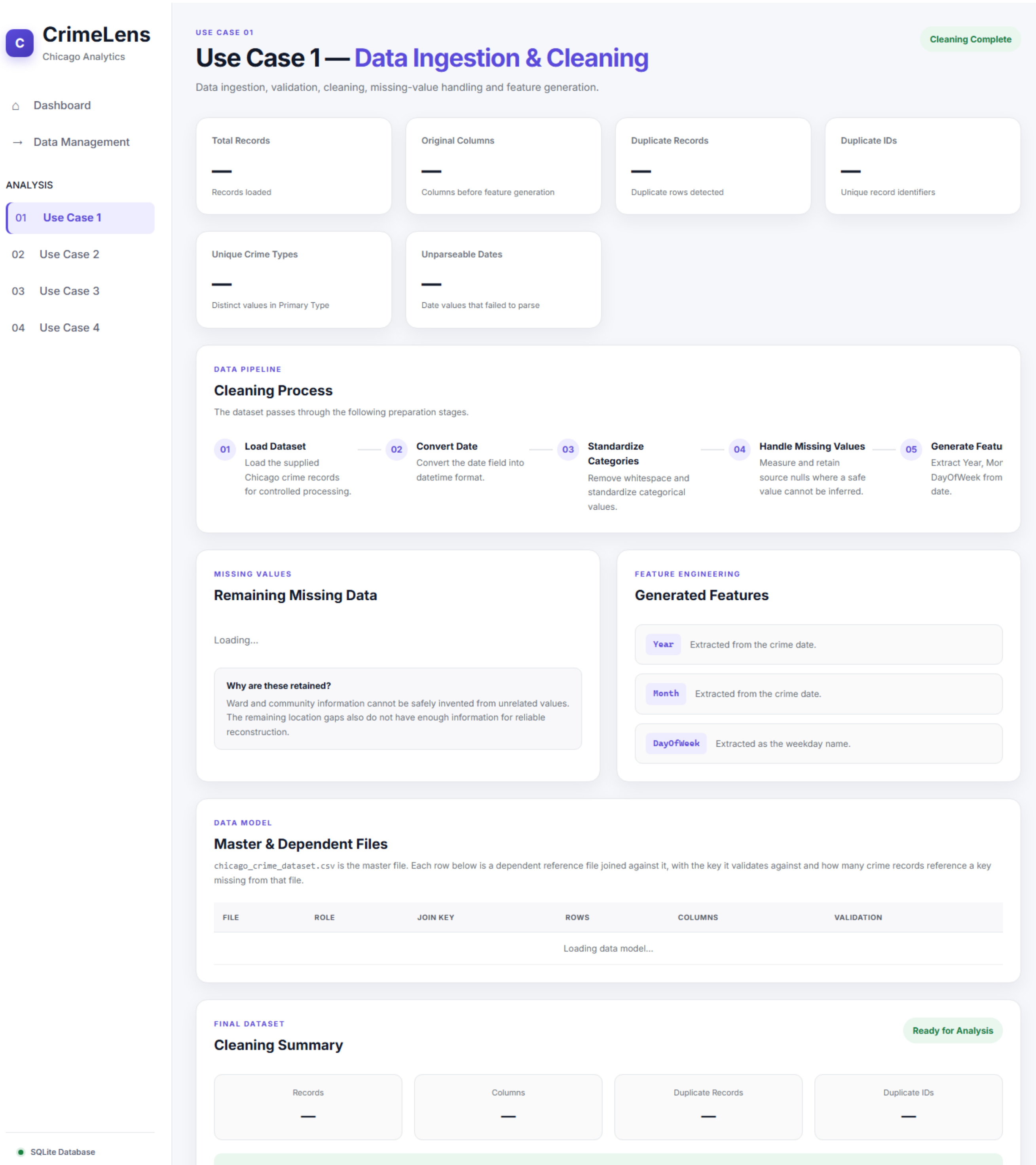
Dashboard



Data Management



Use Case 1


C
CrimeLens
Chicago Analytics
Dashboard
Data Management
ANALYSIS
01 Use Case 1
02 Use Case 2
03 Use Case 3
04 Use Case 4
SQLite Database
USE CASE 01
Use Case 1 — Data Ingestion & Cleaning
Cleaning Complete
Data ingestion, validation, cleaning, missing-value handling and feature generation.
Total Records
—
Records loaded
Original Columns
—
Columns before feature generation
Duplicate Records
—
Duplicate rows detected
Duplicate IDs
—
Unique record identifiers
Unique Crime Types
—
Distinct values in Primary Type
Unparseable Dates
—
Date values that failed to parse
DATA PIPELINE
Cleaning Process
The dataset passes through the following preparation stages.
01 Load Dataset
Load the supplied Chicago crime records for controlled processing.
02 Convert Date
Convert the date field into datetime format.
03 Standardize Categories
Remove whitespace and standardize categorical values.
04 Handle Missing Values
Measure and retain source nulls where a safe value cannot be inferred.
05 Generate Featu
Extract Year, Mor DayOfWeek from date.
MISSING VALUES
Remaining Missing Data
Loading...
Why are these retained?
Ward and community information cannot be safely invented from unrelated values. The remaining location gaps also do not have enough information for reliable reconstruction.
FEATURE ENGINEERING
Generated Features
Year Extracted from the crime date.
Month Extracted from the crime date.
DayOfWeek Extracted as the weekday name.
DATA MODEL
Master & Dependent Files
chicago_crime_dataset.csv is the master file. Each row below is a dependent reference file joined against it, with the key it validates against and how many crime records reference a key missing from that file.
FILE
ROLE
JOIN KEY
ROWS
COLUMNS
VALIDATION
Loading data model...
FINAL DATASET
Cleaning Summary
Ready for Analysis
Records
—
Columns
—
Duplicate Records
—
Duplicate IDs
—
UC1 completed successfully

Use Case 2

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- ⌂ Dashboard
- → Data Management

ANALYSIS

- 01 Use Case 1
- 02 **Use Case 2**
- 03 Use Case 3
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SQLite Database

USE CASE 02

Analysis Ready

Exploratory Analysis

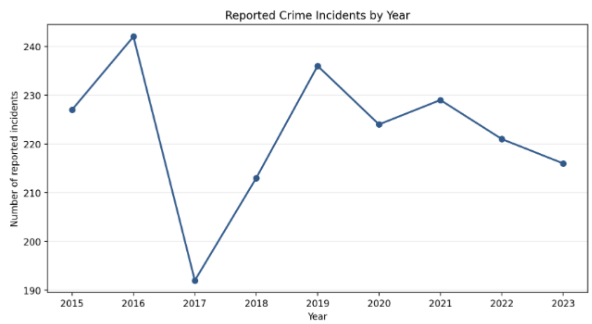
Complete exploratory analysis using the validated Stage 9 outputs.

Total Records	Top Crime	Arrest Rate	Peak Day
2,000	**THEFT**	**29.80%**	**Month 10**
Records in the crime dataset	Most frequent crime type	Percentage of records arrested	Highest crime frequency

TREND ANALYSIS

Crimes by Year

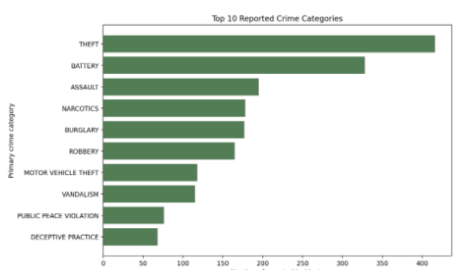
Identify changes in crime volume over time.



CRIME DISTRIBUTION

Top Crime Types

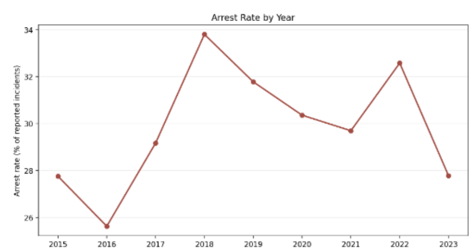
Most frequently occurring crime categories.



ARREST ANALYSIS

Arrest vs No Arrest

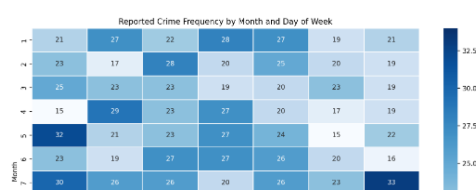
Compare enforcement outcomes.



TEMPORAL ANALYSIS

Month × Day of Week Heatmap

Compare crime frequency across months and weekdays.



Use Case 3

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⌂ Dashboard

→ Data Management

ANALYSIS

01 Use Case 1

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SQLite Database

USE CASE 03

Statistical Analysis

Statistical Insights

Identify statistical patterns, relationships, crime concentrations and potential anomalies in the dataset.

CORRELATION ANALYSIS

Correlation Heatmap

Correlation coefficients across available geographic and administrative fields.

	LATITUDE	LONGITUDE	X_COORDINATE	Y_COORDINATE	WARD_NO	COMMUNITY_CODE
LATITUDE	1.000	-0.013	-0.006	0.013	0.011	0.008
LONGITUDE	-0.013	1.000	0.000	-0.038	-0.036	-0.023
X_COORDINATE	-0.006	0.000	1.000	-0.023	0.002	0.017
Y_COORDINATE	0.013	-0.038	-0.023	1.000	0.010	0.032
WARD_NO	0.011	-0.036	0.002	0.010	1.000	0.039
COMMUNITY_CODE	0.008	-0.023	0.017	0.032	0.039	1.000

Total Crimes
2,000
Records analysed

Unique Crime Types
16
Distinct primary crime categories

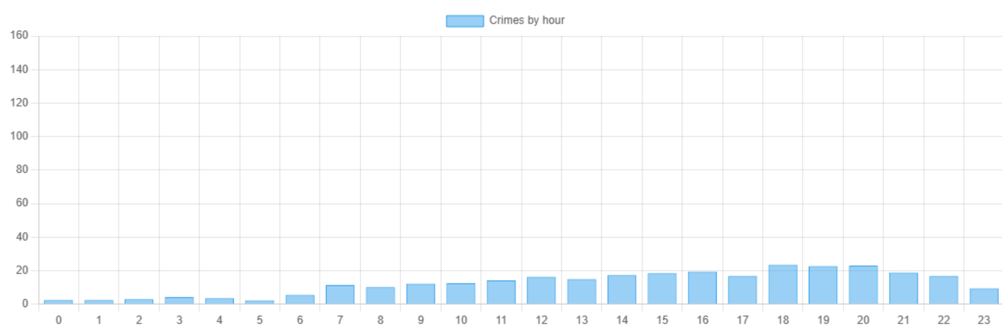
Average Monthly Crimes
166.67
Average crime volume per month

Potential Anomalies
1
Statistical outliers detected

DISTRIBUTION

Crime Intensity by Hour

Identify the hours with the highest recorded crime intensity.



CONCENTRATION

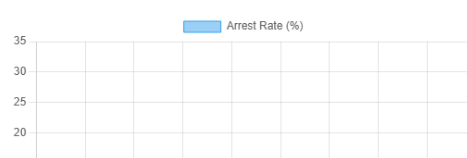
Crime Concentration

Distribution of records among major crime types.

OUTCOME ANALYSIS

Arrest Rate by Crime Type

Compare arrest percentages across major crime categories.



Use Case 4

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- Data Management

ANALYSIS

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SQLite Database

USE CASE 04

Database Reporting

SQLite Connected

Download Use Case 4 Report

Query the cleaned Chicago crime data stored in SQLite and expose analytical results through the REST API.

Database	Crime Records	Crime Types	API Status
SQLite	**2,000**	**16**	**Active**
Local relational database	Records stored	Unique categories	REST endpoints available

REST API

Available Endpoints

The application uses REST endpoints to communicate between the frontend, backend and SQLite database.

GET	`/api/crimes`	Retrieve crime records	Test
GET	`/api/crimes/stats`	Retrieve crime statistics	Test
GET	`/api/crimes/search`	Search crime records	Test
POST	`/api/crimes`	Add a crime record	CRUD
PUT	`/api/crimes/<id>`	Update a crime record	CRUD
DELETE	`/api/crimes/<id>`	Delete a crime record	CRUD

DATABASE REPORTING

Crime Reports

Common analytical queries executed against SQLite.

REPORT 01

Most Common Crimes

Run Query

CRIME_COUNT	PERCENTAGE	PRIMARY_TYPE
416	20.8	THEFT
328	16.4	BATTERY
195	9.75	ASSAULT
178	8.9	NARCOTICS
177	8.85	BURGLARY

REPORT 02

Crimes by District

Run Query

ARREST_COUNT	CRIME_COUNT	CRIME_YEAR
63	227	2015
62	242	2016
56	192	2017
72	213	2018
75	236	2019
68	224	2020
68	229	2021
72	221	2022
60	216	2023